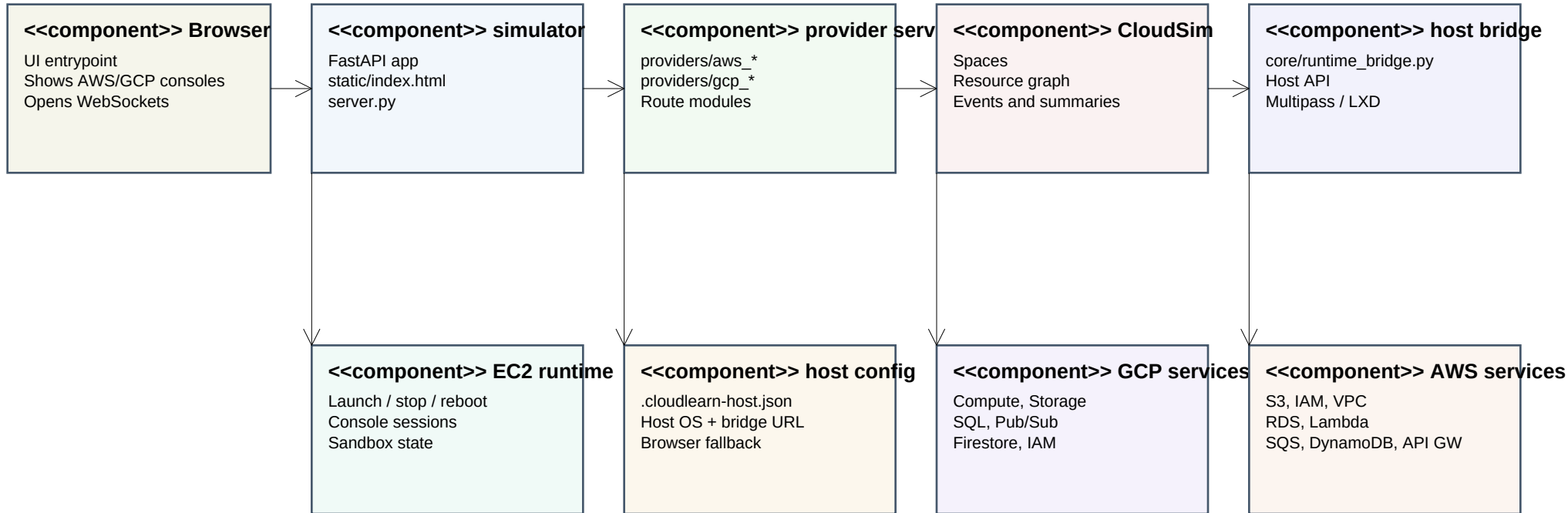


CloudLearn UML Component Diagram

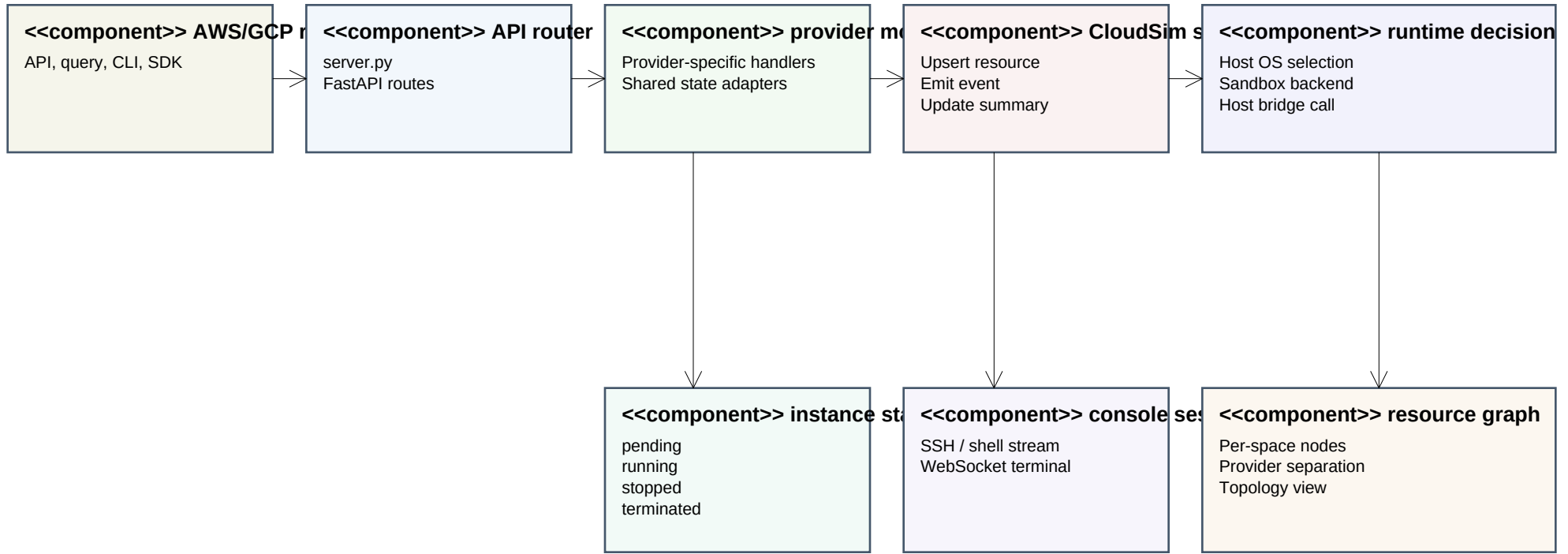
How the simulator, CloudSim, provider services, and host runtime bridge fit together



The browser only talks to the simulator container. The simulator container talks to CloudSim for simulation state and to the host bridge for real host-side runtime control.

How Requests Move

The provider routes are thin; the shared backend does the heavy lifting and mirrors resource changes into CloudSim



CloudSim never launches sandboxes itself. It records and reflects the resource lifecycle, while the host bridge owns the actual runtime execution.

Integration Responsibilities

This keeps each component focused and makes the system easier to reason about

<<component>> simulator container <<component>> CloudSim container <<component>> Host OS bridge

- Serve the browser UI and provider consoles.
- Expose AWS/GCP REST, query, and action routes.
- Maintain shared simulator state and per-space data.
- Forward resource updates into CloudSim.
- Open terminal sessions for running EC2 instances.
- Store active-space state and resource graph summaries.
- Track events, counts, and runtime bundle usage.
- Surface the local cloud backbone for the UI.
- Remain provider-neutral while reflecting provider-requests.
- Execute Multipass on macOS and Windows.
- Execute LXD on Linux.
- Provide bridge health and host status endpoints.
- Keep sandbox launch outside the container boundary.

Important contract

The user-facing browser only talks to the simulator container. The simulator container consumes host OS hints from the launcher config file and uses the host runtime bridge to start the sandbox on the host machine. CloudSim is the simulation backbone that receives resource updates and exposes the active-space graph back to the UI.